

Chapter 6.4

Solutions to Linear Systems Using Matrices

MATH 1710 - College Algebra

Outline

Representing Systems of Linear Equations with Matrices

Representing Systems of Linear Equations with Matrices

What are matrices (singular: matrix):

Definitions:

- **Matrix:** an array of numbers arranged in a rectangular fashion
- **Entry/element:** the numbers in the matrix
- **Dimension:** the number of rows and columns in a matrix ($m \times n$)
- **Square matrix:** a matrix with equal numbers of rows and columns ($n \times n$)

$$\begin{matrix} \begin{bmatrix} 4 & -7 \\ -2 & 9 \end{bmatrix} & \begin{bmatrix} -1 & -5 & 3 \\ 1.2 & 0 & -1.3 \\ 4.1 & 5 & 7 \end{bmatrix} & \begin{bmatrix} -3 & -6 & 9 & 5 \\ \sqrt{2} & -8 & -8 & 0 \\ 3 & 0 & 19 & -7 \\ -11 & -3 & 7 & 8 \end{bmatrix} & \begin{bmatrix} 5 & -2 \\ -2 & \pi \\ 1 & -1 \end{bmatrix} & \begin{bmatrix} 1 & -0.5 & 9 \\ 5 & 0.4 & -3 \end{bmatrix} \\ 2 \times 2 & 3 \times 3 & 4 \times 4 & 3 \times 2 & 2 \times 3 \end{matrix}$$

Figure: Source: our textbook

Representing Systems of Linear Equations with Matrices

See the Concept: Representing a Linear System with a Matrix

A

System of Three Equations

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

B

Coefficient Matrix

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

C

Augmented Matrix

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array} \right]$$

- A** The a_k, b_k, c_k , and d_k are constants and x, y , and z are variables.
- B** The coefficients of the variables are represented in a square matrix called the **coefficient matrix** of the linear system.
- C** The matrix is enlarged to include the constants d_k . The vertical line in this matrix corresponds to where the equals sign occurs in each equation. This matrix is commonly called an **augmented matrix**.

Figure: Source: our textbook

Representing Systems of Linear Equations with Matrices

Examples of representing a system with a matrix:

$$\begin{array}{rcl} 3x - 4y & = & 6 \\ -5x + y & = & -5 \end{array}$$

$$\left[\begin{array}{cc|c} 3 & -4 & 6 \\ -5 & 1 & -5 \end{array} \right]$$

$$\begin{array}{rcl} 2x - 5y + 6z & = & -3 \\ 3x + 7y - 3z & = & 8 \\ x + 7y & = & 5 \end{array}$$

$$\left[\begin{array}{ccc|c} 2 & -5 & 6 & -3 \\ 3 & 7 & -3 & 8 \\ 1 & 7 & 0 & 5 \end{array} \right]$$

Figure: Source: our textbook

Row-Echelon Form

Row-Echelon Form

What is row-echelon form:

Row-echelon form is way of writing a matrix which makes solving the system that it represents simple and easy

Properties:

- The first non-zero entry in each row of the matrix is 1
- The main diagonal has only 1's and 0's

Examples of row-echelon form (main diagonal highlighted blue):

$$\begin{bmatrix} \mathbf{1} & 3 & 0 & -1 \\ 0 & \mathbf{1} & -6 & 1 \\ 0 & 0 & \mathbf{1} & -2 \end{bmatrix} \begin{bmatrix} \mathbf{1} & 2 & 0 \\ 0 & \mathbf{1} & 4 \end{bmatrix} \begin{bmatrix} \mathbf{1} & 3 & -1 & 5 \\ 0 & \mathbf{1} & -1 & 3 \\ 0 & 0 & \mathbf{1} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{1} & 3 & -1 & 5 \\ 0 & \mathbf{0} & 1 & 3 \\ 0 & 0 & \mathbf{0} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{1} & 3 & 5 \\ 0 & \mathbf{0} & 1 \end{bmatrix}$$

Figure: Source: our textbook

Row-Echelon Form

Back-substitution using row-echelon form:

Back-substitution is an easy way of solving a linear system written row-echelon form

Steps:

- The bottom row gives us the value of z
- Plug z into the equation represented by the row above to get y
- Plug y, z into the equation represented by the row above to get x

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 12 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 5 & 5 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Figure: Source: our textbook

Gaussian Elimination

Gaussian Elimination

Obviously, not every system/matrix is given to us already in row-echelon form; we must find it ourselves

Gaussian elimination:

Gaussian elimination is the technique for taking a system/matrix which is NOT in row-echelon form and converting it into row-echelon form

This is done by applying **row transformations** to the matrix which are identical to performing the elimination method on the system of equations

Allowed transformation:

- Any two rows can be interchanged
- Entries on any row can be multiplied by a non-zero constant
- Any row may be altered by adding/subtracting a multiple of another row to/from it

Gaussian Elimination

Gaussian elimination:

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- Any two rows can be interchanged
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Here is an example

Linear System

$$\begin{aligned}2x + 4y + 4z &= 4 \\ x + 3y + z &= 4 \\ -x + 3y + 2z &= -1\end{aligned}$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & 4 & 4 & 4 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array} \right]$$

Figure: Source: our textbook

Row-Echelon Form

Gaussian elimination:

Allowed transformation:

- Any two rows can be interchanged
- Entries on any row can be multiplied by a non-zero constant
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Here is an example with NO solutions (also find the augmented matrix)

$$x - 2y + 3z = 2$$

$$2x + 3y + 2z = 7$$

$$4x - y + 8z = 8$$

Figure: Source: our textbook